Dear Editor and Reviewers,

Thank you for your kind letter dated June 14th. We were pleased to know that our work, recommended by CAD/Graphics 2025, was considered as potentially acceptable for publication in the Journal of Computer & Graphics, subject to adequate revision., subject to adequate revision. We thank the reviewers for the time and effort that they have put into reviewing the manuscript. Their suggestions have enabled us to improve our work. Please find our point-by-point response to the comments raised by the reviewers in below.

Best regards
Sincerely yours
First Author: Wenhu Qin
Corresponding author: Libo Sun

To Reviewer #1:

Thank you very much for your positive evaluation of our work and your thoughtful comments. Your encouraging feedback is truly appreciated. We also sincerely thank you for pointing out areas for further improvement.
Below, we provide detailed responses to your comments:

1. While the method performs well on the tested datasets, further validation on more diverse datasets (e.g., varying lighting, occlusions, ethnicities) would strengthen its robustness claims.

***Response*:** Thank you for this valuable suggestion. In fact, the datasets used in our study already include a wide variety of environmental conditions and subject attributes. For instance, as illustrated in Figure 4 on Page 11 of the revised manuscript, the YawDD dataset contains participants of diverse ethnic backgrounds, while the VBDDD dataset includes numerous samples captured under varying illumination conditions. However, due to privacy restrictions, we are unable to provide more representative visual examples of certain facial features. (*Please see the texts highlighted in yellow in Section 4.1 on Pages 10-11 in the revised manuscript*)

2. Discussion on limitations is brief; a deeper discussion on failure cases or scenarios where the method underperforms would be valuable.

***Response*:** Thank you for highlighting this important point. We have added a discussion of the failure cases in the Conclusion section of the revised manuscript. In particular, we discuss how the performance of our method may degrade under challenging conditions such as severe occlusion and motion blur, which can negatively impact the quality of the extracted heatmaps and downstream classification accuracy. These limitations are now explicitly mentioned in the highlighted text for better transparency. (*Please refer to the texts highlighted in yellow in Section 5 on Page 24 in the revised manuscript*)

To Reviewer #3:

We sincerely thank the reviewer for your encouraging and detailed comments. We are pleased that you found our proposed framework to be well-designed. Your recognition of the clarity of our experimental section is greatly appreciated. We also thank you for your constructive suggestions, which have helped us improve the quality and clarity of our manuscript.
Below, we address each of your comments point by point:

1. The selection of eye and inner-mouth landmarks for stacking is reasonable, but the rationale for excluding other facial landmarks (e.g., eyebrows, face) is not thoroughly justified. Additional experiments exploring different landmark subsets or channel combinations could strengthen the design justification.

***Response*:** Thank you for this insightful comment. To further validate the effectiveness of our landmark selection strategy, we have conducted additional experiments by incrementally stacking other regions like nose and eyebrows, on top of the eyes-mouth region. As shown in Table 1 on Page 14 of the revised manuscript, adding individual regions such as the nose or eyebrows results in negligible or inconsistent performance gains. In contrast, using the full-face landmarks introduces performance degradation across datasets, which we attribute to feature interference from dense, less-informative regions and external visual noise (*e.g., eyeglasses*). These findings confirm that the eye-mouth combination strikes a balance between discriminability and robustness. Additionally, visual comparisons of the corresponding heatmaps before and after training are provided. (*Please see the Figure 5 on Page 13 and the texts highlighted in yellow discussed in Section 4.2 on Pages 12-15 of the revised manuscript*)

2. Aggregating multiple landmarks into a single heatmap may introduce feature interference and reduce the network's ability to distinguish closely spaced landmarks. The paper does not quantify the impact of stacking on localization precision or the sensitivity to the Gaussian spread parameter σ.

***Response*:** We appreciate your pointing out this important omission. In response, we have added a dedicated experiment to analyze the effect of varying the Gaussian spread parameter σ. We find that a small σ value produces sharply localized heatmaps, which enhance spatial precision but are more sensitive to visual noise (*e.g., eyeglasses*). Conversely, large σ values result in overly diffused responses that blur the distinction between adjacent landmarks. These observations and explanations have been added to Section 4.2 of the revised manuscript and are highlighted in blue. (*Please see the Table 2 on Page 16 and the Figure 6 on Page 8 of the revised manuscript*)

3. It remains uncertain whether the method meets real-time requirements on actual vehicle electronic system or low-power processors. The authors should discuss potential deployment challenges under automotive-grade hardware constraints or include additional tests on representative embedded platforms.

***Response*:** Thank you for raising this important concern. We agree that the feasibility of deployment on automotive-grade embedded platforms is essential for practical applications. In our work, we have already conducted latency and memory usage tests on the NVIDIA Jetson AGX Orin, which is one of the most widely used automotive-grade embedded computing platforms in industry today. To further clarify this point, we have added a brief explanation and supporting references to

emphasize the relevance and representativeness of this platform. (*Please see the texts highlighted in yellow in Section 4.5 on Pages 21-23 in the revised manuscript*)

To Reviewer #4:

We sincerely thank the reviewer for your constructive feedback and for recognizing the novelty and practical value of our proposed video-based drowsiness detection framework. Your thoughtful suggestions have helped us improve both the clarity and rigor of the manuscript.
Please find our responses to your comments below:

1. The citations of figures in the paper should use "Fig." or "Figure" instead of "Fig".

***Response*:** Thank you for pointing this out. We have carefully revised all figure citations in the manuscript to consistently use the correct format (“Fig.” or “Figure”) where appropriate.

2.Multiple typos require correction, including but not limited to: Table 3: "Resisual", Section 4.2: "tree drowsiness detection datasets".

***Response*:** We appreciate your careful reading. We have corrected these and other typographical errors throughout the manuscript to improve its presentation quality.

3.The enhancement of geometric relationships through diagonal embedding requires further elaboration.

***Response*:** Thank you for this helpful suggestion. We have expanded the explanation of diagonal embedding in the revised manuscript. Specifically, we clarify that this mechanism modulates the intensity values of the heatmap channels based on spatial location, thereby enriching the response map with implicit geometric context. (*Please see the paragraph highlighted in yellow in Section 3.2 on Page 9 in the revised manuscript.*)

4.The paper states that "the spatial stream has 8 attention heads and the temporal stream has 16 attention heads". Although the effectiveness of the attention mechanism has been proven, is there any specific consideration behind the choice of different numbers of attention heads?

***Response*:** We thank the reviewer for raising this point. The higher number of attention heads in the temporal stream is designed to enhance its capacity for aggregating cues across multiple sampled frames, reducing reliance on any single frame. This helps mitigate the impact of occasional noise such as motion blur or heatmap segmentation failures in individual frames. (*Please see the explanation highlighted in blue in Section 4.1 on Page 12 in the revised manuscript.*)

5.In the paper, real-time performance is mentioned, but the experimental section lacks comparison with other methods despite presenting memory consumption and time delay data on different hardware platforms, making it difficult to directly

evaluate the superiority in real-time performance.

***Response*:** Thank you for pointing this out. To better assess real-time performance, we have added a comparison of total processing delay against several existing video-based driver drowsiness detection methods, using their reported hardware settings. Although direct comparisons across different platforms may not be fully rigorous, the results provide valuable insights into the relative efficiency of our method. Our approach achieves significantly lower latency while running on less powerful hardware, demonstrating strong potential for embedded deployment. (*Please see the texts highlighted in blue in Section 4.5 and Table 7 on Pages 23-24 in the revised manuscript.*)

Video-based Driver Drowsiness Detection using Stacked Landmark Heatmaps and Spatial-temporal Attention

Abstract

Driver drowsiness poses a critical threat to road safety and remains a persistent challenge even in advanced autonomous driving systems. However, existing vision-based detection methods extract facial features inefficiently and are susceptible to cumulative errors from external face alignment models. To address these limitations, we develop a novel video-based framework that combines trainable facial feature extraction with efficient spatial-temporal analysis. Our feature extraction module employs stacked landmark heatmaps to extract structural facial features efficiently while can be trained for drowsiness-specific features. Furthermore, we introduce a diagonal embedding strategy to enhance the geometric relations between landmarks of a single stacked heatmap. To adequately analyze the spatial-temporal features, a two-stream structure combining 2D and 3D convolutions with multi-head self-attention is utilized in this article. Experimental results validate that our method outperforms existing methods and achieves higher detection accuracy.

Keywords: driver drowsiness detection, video processing, face alignment, landmark heatmap, intelligent cabins

1. Introduction

Driver drowsiness is a major factor leading to traffic accidents and poses a serious threat to road safety. With the rapid development of autonomous driving technologies, the role of the driver has gradually shifted from active controller to passive supervisor[1]. Nevertheless, driver state monitoring remains a key technology for ensuring safety even in highly automated driving environments[2]. Research has shown that drivers may be more prone to drowsiness in automated scenarios compared to manual driving[3, 4]. Moreover, core modules in autonomous driving systems, such as perception and planning, consume significant computational resources[5, 6], which

raises constraints for auxiliary modules like drowsiness detection. Thus, developing efficient and lightweight drowsiness detection methods is essential for improving the intelligence and safety of future intelligent cabins.

Currently, many global automotive companies are actively exploring drowsiness detection technologies[7]. According to the difference of the source for obtaining the information, these methods can be divided into three categories[8]: physiological signals like EEG[9, 10, 11], vision-based methods[12, 13, 14], and multi-modal fusion methods[15, 16]. While physiological signals are widely used in laboratories, their inconvenience in real-world deployment limits practical application[17]. In contrast, vision-based methods have gained increasing attention due to advances in computer vision, offering a more feasible and deployable solution.

Vision-based methods generally start with extracting facial features. Common descriptors include ROIs[18], optical flow[19], and facial landmarks[13]. Among them, facial landmarks are particularly suitable for drowsiness detection due to their nature of structural representation to faces. However, these methods often rely on external face alignment models, which are computationally expensive and not trainable within the drowsiness detection pipeline, leading to accumulated errors. After feature extraction, drowsiness is often assessed by analyzing temporal variations in facial behaviors. Metrics like PERCLOS[20] and blink rate[21] are commonly used, but such threshold-based methods suffer from poor adaptability due to individual differences. Some studies utilize neural networks to extract spatial-temporal features from image sequences[22], and recent work has explored two-stream structures that combine spatial and temporal features to better detect drowsiness[23].

Nonetheless, two key limitations remain in vision-based methods: (1) they rely on external face alignment methods that cannot be trained for drowsiness detection, and (2) low efficiency and accuracy in spatial and temporal analysis, which restrict real-time performance in intelligent cabins. To address these issues, we propose a drowsiness detection framework based on stacked landmark heatmaps and a two-stream attention analysis module. Specifically, we use an efficient segmentation model to extract facial features, and employ a diagonal embedding strategy to enhance spatial structural representation, then introduce a two-stream attention module to jointly analyze spatial and temporal features from stacked heatmaps for accurate drowsiness detection.

The rest of this article is organized as follows: Section 2 reviews related

work on driver drowsiness detection, Section 3 details our proposed method, Section 4 presents experiments on public datasets, and Section 5 concludes the article and discusses future directions.

2. Related Work

2.1. Facial Feature Extraction

Facial feature extraction is a crucial step in vision-based driver drowsiness detection. Region of interest (ROI) is a basic strategy, where eyes and mouth region are extracted for analysis. Sun et al.[18] proposed a fusion network (FFF-CNN), including both global facial and local eye ROIs, with feature attention and stream interaction modules to improve robustness under low-quality image conditions. Hu et al.[24] introduced a method integrating visual saliency and gaze estimation, using eye ROIs to coarsely localize gaze regions and refine driver's attention prediction through saliency maps. Some studies further utilized optical flow to capture dynamic changes in facial regions over time. Zhang et al.[25] employed TLD tracking and Kalman filtering for robust nose tracking, then extracted vertical optical flow and gradient cues for yawning detection. Akrout et al.[19] extracted mouth motion vectors via optical flow and used an SVM classifier to recognize yawning. Mou et al.[26] proposed IsoSSL-MoCo, fusing temporal features of eyes, mouth, and head motion with attention to enhance detection.

Facial landmarks have also been widely adopted due to their structural and interpretable nature. Lu et al.[13] proposed JHPFA-Net to combine head pose and geometric features derived from landmark distances and angles to detect yawning. Kim et al.[27] used landmarks for eye closure and head pose analysis to detect drowsiness and distraction. However, many state-of-the-art face alignment methods generate heatmaps and use Soft-argmax[28] to regress coordinates, such as FARL[29]. Most methods still rely on external face alignment models trained for generic localization accuracy, not specific for drowsiness detection. Although Wu et al.[30] proposed YOLOFaceMark , a lightweight face detection and alignment model improving inference efficiency for drowsiness detection, it still relies on traditional geometric metrics like eye aspect ratio (EAR), which limits its ability to extract drowsiness-specific facial features between landmarks.

While existing facial feature extraction methods improve detection interpretability and efficiency, they face limitations. On one hand, ROI and optical flow methods depend on predefined regions and local motion,

making them sensitive to lighting and pose variations, on the other hand, although landmarks have strong structural expressiveness, they rely on face alignment models trained on static image tasks, which are difficult to continue training for temporal tasks like drowsiness detection, thereby limiting their ability to capture extended temporal features.

2.2. Spatial-temporal Feature Analysis

Drowsiness evolves over time, and the dynamics of facial features contain important temporal information. Accordingly, existing research has extended facial analyzed by incorporating temporal sequences from video to better analyze these features.

A common approach is to compute statistical metrics over time windows. For instance, Dasgupta et al.[20] used a lightweight eye state classifier to compute PERCLOS within a sliding window. Hu et al.[21] used brightness and contour to estimate eye closure and derived PERCLOS, blink rate, and duration.

Deep learning methods extract spatial-temporal features from image sequences. Guo et al.[22] combined CNN and LSTM for frame-level feature extraction and temporal modeling. Weng et al.[31] fused eyes, mouth, and head features using DBN-HMM to improve perceiving temporal changes in drowsy behavior. Yang et al.[32] used landmark sequences with 3D CNNs and BiLSTM in a 3D-LTS framework to detect yawning. Lin et al.[33] proposed MAGCN using LSTM, attention, and graph convolution to model region-level temporal dependencies.

Some recent studies have attempted to decouple spatial structure and temporal dynamics using multi-stream structures. Bai et al.[23] proposed 2s-STGCN using landmark sequences to build spatial and temporal graphs for graph convolution. Yang et al.[34] used CSP-selected landmarks and local ROIs, with two-branch attention for spacial-temporal analysis.These methods improve accuracy and robustness but depend on external alignment models for landmark localization, which increases inference latency and may introduce cumulative errors. Yang et al.[34] showed that alignment delays dominate the video processing time.

To address this, we propose a method based on pre-trained stacked heatmap segmentation. It retains landmarks' expressive ability while enabling task-specific fine-tuning within the full drowsiness detection pipeline, achieving faster inference and lower computational cost.

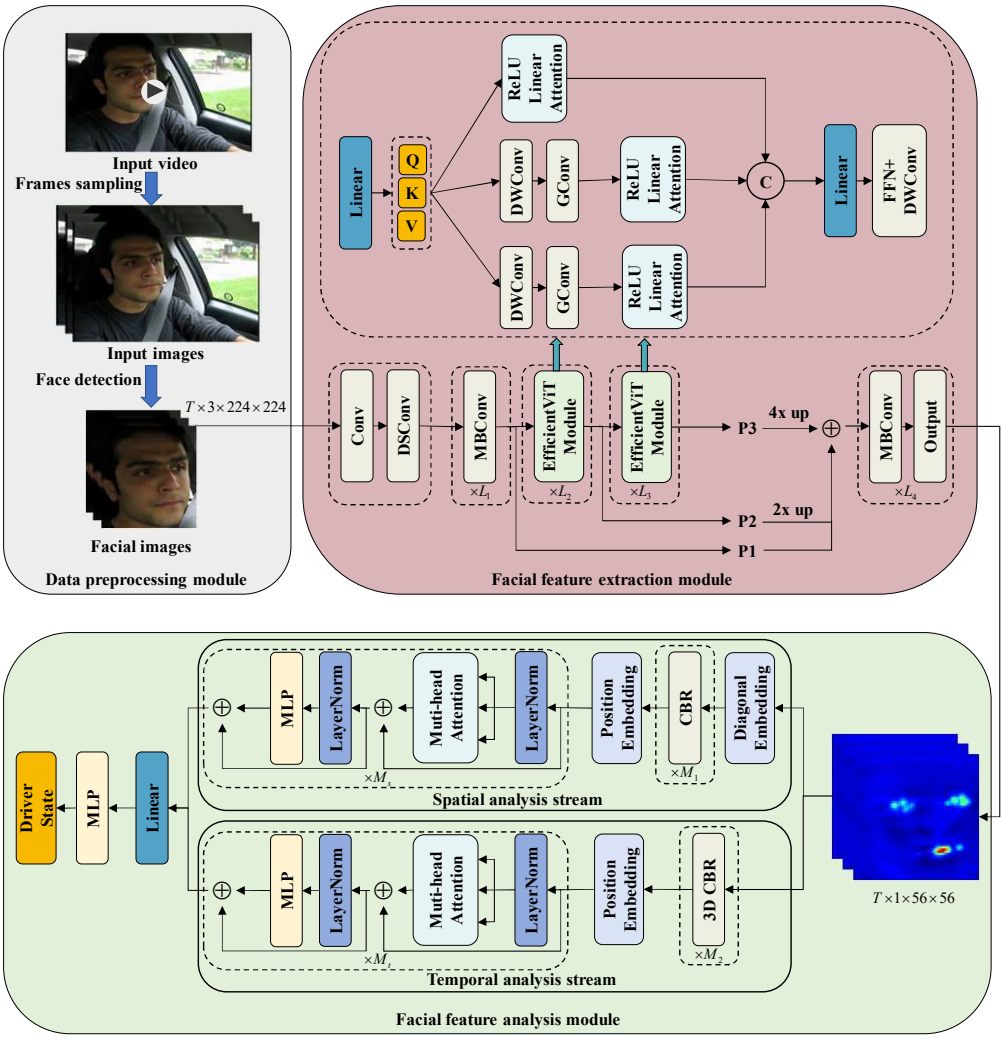


Figure 1: Pipeline of the proposed video-based drowsiness detection algorithm. DSConv(Depthwise Separable Convolution), MBConv(Mobile inverted Bottleneck Convolution), DWConv(Depthwise Convolution), and GConv(Grouped Convolution) are lightweight convolution modules for efficient computation. ReLU Linear Attention is a linear approximation to standard self-attention.

3. Proposed Method

The proposed video-based driver drowsiness detection algorithm based on stacked multi-landmark heatmap segmentation consists of three modules:

data preprocessing, facial feature extraction, and feature analysis, as shown in Fig. 1. In the data preprocessing module, T frames are sampled at equal time intervals from the video, and facial regions are extracted using Retinaface[35] face detection model. These facial images are then input to the facial feature extraction module as a batch.

The facial feature extraction module is a segmentation model based on EfficientViT[36], which outputs T stacked multi-landmark heatmaps from the input facial images. These heatmaps are passed into the feature analysis module.

The facial feature analysis module utilizes a two-stream structure to analyze heatmaps. In the spatial stream, we first apply diagonal embedding to enhance geometric relationships between landmarks. The resulting feature heatmaps are processed through convolution and multi-head self-attention to analyze spatial features. In the temporal stream, 3D convolution and multi-head self-attention are used to analyze temporal features across frames. Finally, spatial and temporal features are fused and passed through an MLP for driver drowsiness detection.

3.1. Facial Feature Segmentation with Stacked Multi-Landmark Heatmaps

Facial alignment is a commonly used method to represent facial structure and is widely used in driver drowsiness detection. However, for an alignment task involving N_p facial landmarks, Soft-argmax[28] regression must be performed $2N_p$ times.This regression operation can be regarded as a convolution with a kernel of the same size as the heatmap, where the x and y coordinates are obtained independently through weighted summation across the entire heatmap. While this is acceptable for single-image tasks requiring precise coordinate prediction, it leads to significant computational waste in video-based drowsiness detection, where T frames need to be processed and exact coordinates are not required. Therefore, we propose stacked multi-landmark heatmaps output by a segmentation model. These heatmaps are used to replace face alignment, which helps reduce computational overhead and better suit the needs of drowsiness detection.

Specifically, during training of our facial feature extraction module, we directly stack the heatmaps of selected landmarks into a heatmap that includes multiple facial regions. Compared to independent per-landmark regression, this method avoids repeated Soft-argmax[28] regression and reduces memory usage by decreasing the number of heatmaps. Moreover, the stacked heatmaps represent facial regions collectively and maintain the same facial feature

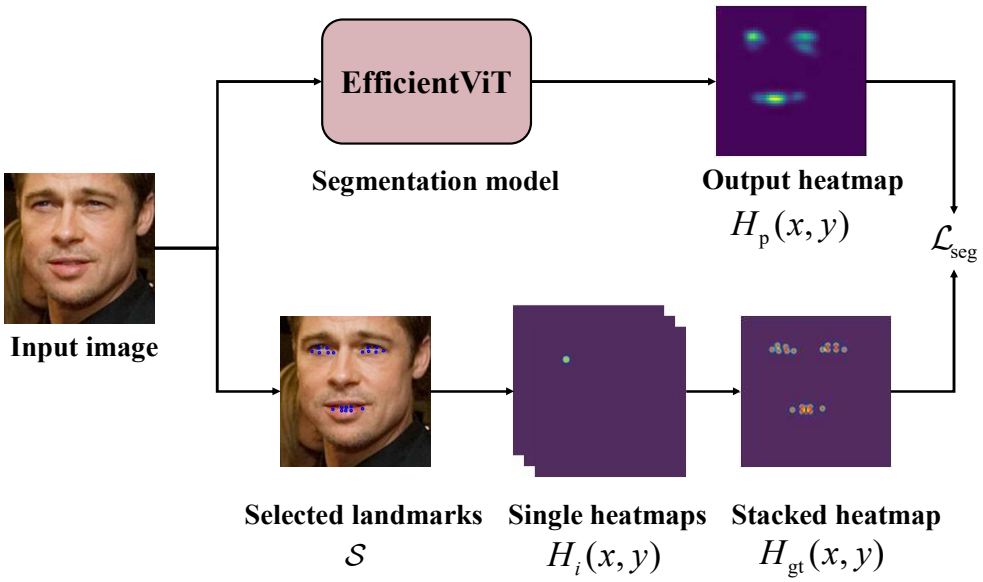


Figure 2: The training process of the stacked heatmap segmentation model.

expression capacity as traditional landmarks. The training process is shown in Fig. 2.

Given a dataset with N_p landmarks and annotated coordinates $(x_i, y_i)_{i=1}^{N_p}$, each heatmap is generated using two-dimensional Gaussian distributions:

$$H_i(x, y) = \exp\left(-\frac{(x - x_i)^2 + (y - y_i)^2}{2\sigma^2}\right) \tag{1}$$

where σ is the standard deviation of the Gaussian, controlling the expansion of heatmap. In this article, we set the default value of σ to 1. Then, the heatmaps for a selected subset $\mathcal{S}$ (eye and inner mouth landmarks) are stacked:

$$H_s(x, y) = \sum_{i \in \mathcal{S}} H_i(x, y) \tag{2}$$

The result is normalized and resized to match the segmentation model output. The selected subset $\mathcal{S}$ only includes facial landmarks from regions of eyes and inner mouth to ensure focus on areas correlated with drowsiness, while avoiding interference caused by excessive stacking

The model is trained using L1 loss:

$$\mathcal{L}_{seg} = \frac{1}{n} \sum |H_p(x, y) - H_{gt}(x, y)| \tag{3}$$

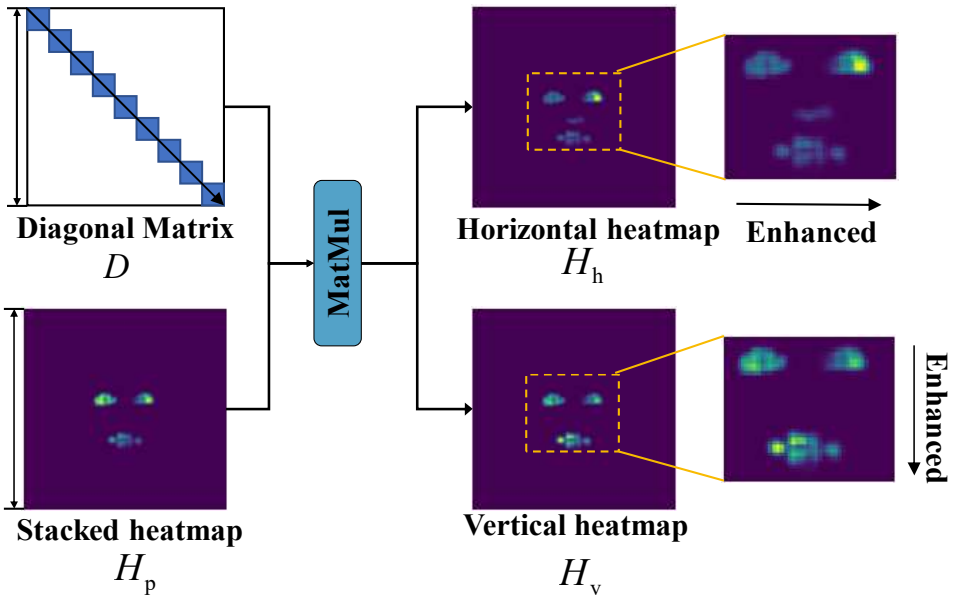


Figure 3: The process of diagonal embedding, the diagonal matrix and the stacked heatmap have the same dimensions.

where n is the number of pixels per heatmap, H_p denotes the predicted heatmap output by the segmentation model, and H_{gt} is the corresponding ground truth heatmap.

Through pre-training, the proposed stacked heatmap segmentation model not only effectively preserves the structural features of key facial regions related to drowsiness detection but also achieves lower computational resource consumption and inference delay, providing structured facial features for subsequent drowsiness detection.

3.2. Diagonal embedding

Although the stacked multi-landmark heatmap enables efficient facial feature extraction, the direct stacking process during training overlooks the spatial relationships among landmarks, resulting in limitations in spatial representation. To address this, we further draw inspiration from the idea of matrix-weighted heatmap regression in Soft-argmax[28], and propose diagonal embedding to enhance spatial geometric relationships on the existing stacked heatmaps. As shown in Fig. 3, let the stacked heatmap output from the

segmentation model be $H_{\mathrm{p}} \in \mathbb{R}^{h \times w}$, where h and w are the height and width of the heatmap, respectively. In this article, $h = w$. We apply horizontal and vertical position weighting using a diagonal matrix $D \in \mathbb{R}^{h \times h}$, where:

$$D(i, i) = i, 0 \leq i < w \tag{4}$$

The spatial features can be embedded by:

$$\begin{cases} H_v = D \cdot H_p \\ H_h = H_p \cdot D \end{cases} \tag{5}$$

In this formulation, H_v represents a vertically embedded heatmap where each row is weighted proportionally to its vertical position, while H_h encodes horizontal positional cues through column-wise weighting. Note that this operation does not change the spatial position of the heatmap pixels, but modulates their intensity values based on spatial location, thereby enriching the response map with implicit geometric context.

This diagonal embedding enables the feature analysis module to better capture structural relationships among facial regions, such as the degree of eye closure and mouth opening, which are critical for drowsiness-specific feature analysis, by making the heatmap values more sensitive to their absolute spatial coordinates.

3.3. Spatial-Temporal Two-Stream Feature Analysis

To analyze spatial-temporal features of facial heatmaps, we propose a spatial-temporal two-stream analysis module, as shown in Fig. 1. The input is a batch of T stacked heatmaps. Compared with the original images batch size $T \times 3 \times 224 \times 224$, the heatmaps batch size $T \times 1 \times 56 \times 56$ enables more efficient processing. Therefore, a residual multi-head self-attention module is adopted for simplicity, instead of the more efficient EfficientViT[36] block used in the feature extraction module.

In the spatial stream, we apply diagonal embedding to enhance spatial structure. Then, $2 \times T$ heatmaps are concatenated along the channel dimension and passed through a CBR (Conv-BN-ReLU) module for further dimensionality reduction. Multi-head self-attention is then applied to analyze spatial relations in single heatmap.

In the temporal stream, 3D CBR is applied to extract temporal features of T heatmaps. Multi-head self-attention is used again to reinforce temporal

dependencies across frames. The final classification output c_p is obtained by feeding the concatenated spatial and temporal embeddings $[E_s, E_t]$ into an MLP classifier:

$$c_p = \text{MLP}([E_s, E_t]) \tag{6}$$

where E_s and E_t denote the spatial and temporal feature embeddings, respectively. The classification loss is computed using a BCE loss between the predicted drowsiness score c_p and the ground truth label c_{gt}:

$$\mathcal{L}_{\text{cls}} = [c_{gt} \cdot \log(c_p) + (1 - c_{gt}) \cdot \log(1 - c_p)] \tag{7}$$

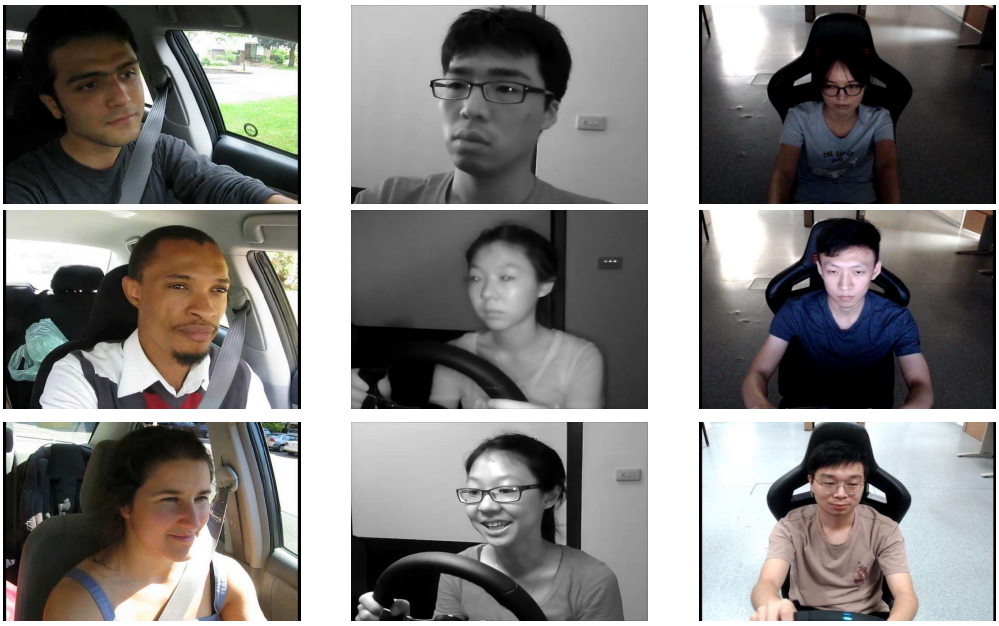
To ensure effective facial representation, we jointly train the segmentation model and analysis modules with different learning rates. We limited the segmentation model's learning rate in drowsiness detection training, to preserve region extraction while adapting to drowsiness-specific features.

4. Results

4.1. Datasets and implemental details

We pre-trained the heatmap segmentation model on the 300W[37] dataset, which contains 3,148 training images and 689 testing images. Each image is labeled with 68 facial landmarks, covering 600 face instances across indoor and outdoor scenarios.

We evaluated our method on three popular driver drowsiness datasets: YawDD[38], NTHU-DDD[31], and VBDDD[34]. For YawDD[38] and NTHU-DDD[31], we re-labeled videos based on original labels to segment standardized samples, following the general data processing in video-based drowsiness detection[23, 26, 34]. While for VBDDD[34], we directly use the original unsegmented videos to ensure fair comparison. YawDD[38] consists of 342 videos from 107 drivers, covering normal driving, talking, and yawning behaviors. This dataset includes participants from multiple ethnic backgrounds and was collected under naturalistic driving conditions, as shown in Fig. 4(a). We re-labeled and segment these labeled as yawning and normal into 579 clips, which were split into 312 training and 267 validation samples. NTHU-DDD[31] provides 450 videos recorded from 36 participants across five scenarios including bare face, glasses and night conditions. The dataset also contains subjects of diverse ethnicities and includes both daytime and nighttime recordings captured with an infrared camera, as shown in Fig. 4(b). Following the official dataset split, we used videos from 18 participants for



(a) YawDD **(b) NTHU-DDD** **(c) VBDDD**

Figure 4: Example frames from the three drowsiness detection datasets.

training and 4 for validation. All videos include frame-level drowsy and normal labels, and we further segmented them into 1,071 single-state clips for analysis. VBDDD[34] includes 558 videos from 37 participants and was explicitly designed to include three lighting conditions(dim, normal and bright), allowing controlled evaluation of model performance under varying illumination, as shown in Fig. 4(c). We randomly split the original videos into 502 training and 56 validation samples without additional processing. All video samples used in our experiments range from 5 to 50 seconds in duration.

In this study, we set the default number of sampled frames to $T = 16$, following standard protocols in video classiffication task established by Wang et al.[39]. The heatmap segmentation model is based on EfficientViT-Seg-L2[36], with a modified segmentation head outputting a tensor of shape $1 \times 56 \times 56$. During pre-training, the initial learning rate was set to 1×10^{-4} with a total of 200 epochs. The final model achieved an average L1 loss of approximately 0.004 on the validation set, and further training did not achieve noticeable

improvements, we conclude that the model has been adequately trained.

The feature analysis network adopts a two-stream structure implemented by three CBR modules and three 3D CBR modules. Residual multi-head self-attention is applied, with 8 heads in the spatial stream and 16 in the temporal stream. The higher number of attention heads in the temporal stream is designed to enhance its capacity for aggregating cues across the sampled frames, rather than relying too heavily on any single frame. This helps mitigate the impact of unreliable heatmaps caused by occasional issues such as motion blur or segment failure in individual frames. In drowsiness detection experiments, the training phase uses an initial learning rate of 1×10^{-4} for 30 epochs, the pretrained segmentation model is trained with a limited learning rate of $0.01\times$ the base rate.

The Adam optimizer is used with parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$. All training is conducted on an NVIDIA GeForce RTX 3090 GPU using PyTorch 2.2.1 and CUDA 12.1. In the inference stage, we deploy the model on an NVIDIA Jetson AGX Orin in-vehicle embedded platform with PyTorch 1.13.1 and CUDA 11.4, while other hardware platforms maintain software consistency with the training environment.

4.2. Evaluation

To comprehensively evaluate the design choices behind the heatmap-based representation, we conduct experiments on three drowsiness detection datasets. First, we systematically vary the facial regions used for heatmap stacking during pretraining to assess the contribution of different landmark subsets. Since each heatmap is generated by mapping landmark coordinates through a two-dimensional Gaussian, we further investigate the impact of the Gaussian spread parameter σ on the quality and discriminability of the stacked heatmaps. Finally, we explore different fine-tuning strategies applied to the segmentation model in the downstream task to understand how learning rate affects classification performance.

By selectively excluding specific facial regions, we analyze their contribution to drowsiness detection. Fig. 5 shows that fine-tuning improves the model's ability to extract task-specific features. For example, when the eye region is included, surrounding heatmap responses are more suppressed, indicating clearer separation of landmarks. Similarly, task-driven training in YawDD[38] prioritizes relevant regions. After fine-tuning, the heatmap intensity decreases around the eyes and shifts focus to the mouth, as it is a stronger indicator of yawning.

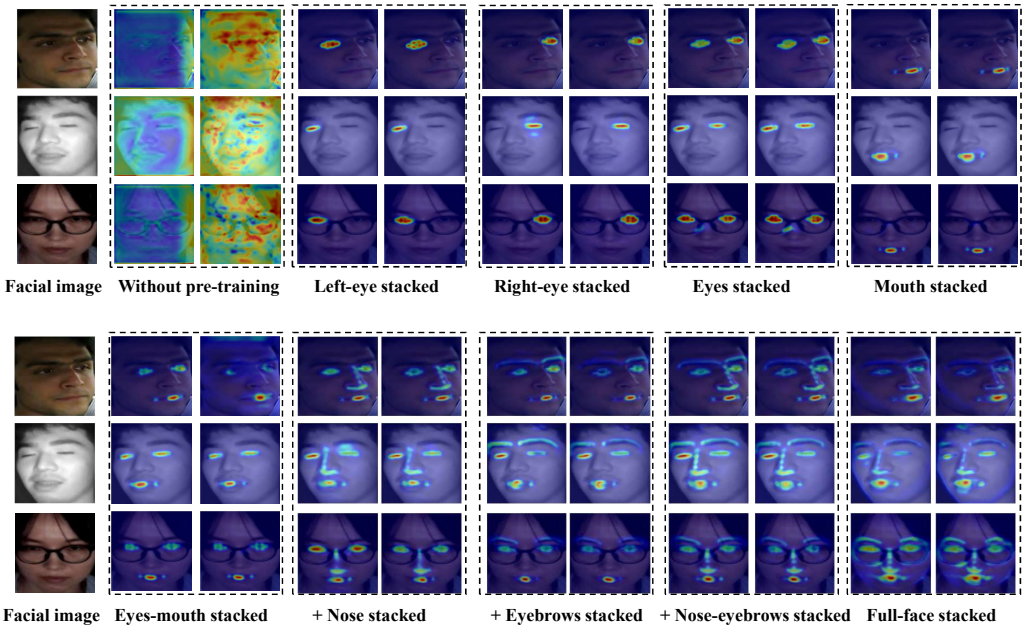


Figure 5: Comparison of facial feature heatmaps before and after drowsiness detection training under different stacking regions, the three rows of images are from three datasets, with the left side of each column showing pre-training results and the right side showing post-training results. Here, symbol "+" indicates additional regions stacked on top of the eyes-mouth combination, and "Full-face" indicates that all 68 landmarks are stacked. For clear visualization, all heatmaps in this figure and subsequent ones are overlaid on facial images after transparency adjustment.

Table 1 shows that exclude any drowsiness-specific facial region negatively impacts final detection accuracy, though the extent varies across datasets and tasks. In YawDD[38], which targets yawning behavior, excluding the mouth region leads to the most significant drop. In NTHU-DDD[31], missing the eye region results in the greatest performance loss. In VBDDD[34], excluding either the eye or mouth region produces comparable degradation.

Table 1: Classification performance under different facial heatmap stacking strategies across three datasets. Here, symbol "+" indicates additional regions stacked on top of the eyes-mouth combination, and "Full-face" indicates that all 68 landmarks are stacked. The best results for each dataset are marked in bold.

Dataset	Region	Prec.	Rec.	F1	Acc
YawDD[38]	-	0.7377	0.6966	0.7023	0.7292
	Left Eye	0.8404	0.7943	0.8040	0.8229
	Right Eye	0.7976	0.7869	0.7903	0.8021
	Eyes	0.8846	0.8487	0.8589	0.8698
	Mouth	0.9510	0.9412	0.9450	0.9479
	Eyes-mouth	**0.9904**	**0.9935**	**0.9919**	**0.9922**
	+ Nose	0.9610	0.9577	0.9588	0.9604
	+ Eyebrows	0.9486	0.9438	0.9505	0.9521
	+ Nose-eyebrows	0.9577	0.9485	0.9433	0.9604
	Full-face	0.8746	0.8703	0.8710	0.8726
NTHU-DDD[31]	-	0.6417	0.6411	0.6356	0.6790
	Left Eye	0.8658	0.8414	0.8458	0.8488
	Right Eye	0.8774	0.8842	0.8793	0.8889
	Eyes	0.8541	0.8756	0.8597	0.8673
	Mouth	0.7996	0.8090	0.8012	0.8149
	Eyes-mouth	0.9444	0.9331	0.9383	0.9444
	+ Nose	0.9241	0.9380	0.9301	0.9351
	+ Eyebrows	**0.9457**	**0.9421**	**0.9439**	**0.9491**
	+ Nose-eyebrows	0.8894	0.9070	0.8962	0.9074
	Full-face	0.8841	0.9094	0.8924	0.8981
VBDDD[34]	-	0.6169	0.5764	0.5605	0.6250
	Left Eye	0.8048	0.7872	0.7934	0.8036
	Right Eye	0.7871	0.7612	0.7626	0.7768
	Eyes	0.8581	0.8342	0.8359	0.8512
	Mouth	0.8707	0.8465	0.8542	0.8571
	Eyes-mouth	**0.9423**	**0.9545**	**0.9456**	**0.9464**
	+ Nose	0.9239	0.9328	0.9271	0.9286
	+ Eyebrows	0.9064	0.9111	0.9093	0.9107
	+ Nose-eyebrows	0.8902	0.8985	0.8933	0.8983
	Full-face	0.8883	0.8719	0.8779	0.8750

Furthermore, while incremental stacking of additional regions such as the nose and eyebrows introduces slight variations in performance, Table 1 clearly shows that the eyes-mouth combination yields near-optimal results across all datasets. Although adding individual regions such as the nose or eyebrows generally brings no consistent improvement, NTHU-DDD[31] exhibits a slight performance boost when eyebrows are included. This suggests that the additional information they contribute may be redundant or less discriminative for the drowsiness detection task. In contrast, stacking both the nose and eyebrows or using the full face tends to reduce performance. We attribute this to feature interference introduced by densely distributed landmarks and the inclusion of less informative or distracting areas. In particular, full-face heatmaps may be affected by external factors such as eyeglasses, as shown in Fig. 5, which distort the heatmap value of the eye region and mislead the model during feature analysis.

Since all heatmaps in our method are generated by applying a two-dimensional Gaussian centered on each landmark, it is also essential to analyze how the spread parameter σ influences the segmentation quality and subsequent classification performance, the results are shown in Table 2.

Table 2: Classification performance under different Gaussian spread parameters σ across three datasets. The best results for each dataset are marked in bold.

Dataset	σ	Prec.	Rec.	F1	Acc
YawDD[38]	0.1	0.7846	0.7803	0.7810	0.7826
	0.5	0.8761	0.8631	0.8643	0.8674
	1	**0.9904**	**0.9935**	**0.9919**	**0.9922**
	2	0.9320	0.9169	0.9126	0.9253
	5	0.8195	0.8042	0.7989	0.8131
NTHU -DDD [31]	0.1	0.8755	0.9051	0.8821	0.8878
	0.5	0.9030	0.9226	0.9116	0.9165
	1	**0.9444**	**0.9331**	**0.9383**	**0.9444**
	2	0.8999	0.9077	0.9062	0.9121
	5	0.8550	0.8699	0.8610	0.8704
VBDDD[34]	0.1	0.8520	0.8482	0.8446	0.8482
	0.5	0.8705	0.8893	0.8730	0.8929
	1	**0.9423**	**0.9545**	**0.9456**	**0.9464**
	2	0.8946	0.9124	0.9050	0.9094
	5	0.8962	0.8778	0.8836	0.8889

As shown in Fig. 6, a larger sigma causes the Gaussian responses of stacked landmarks to overlap substantially, forming a broad circular region that undermines the structured representation of landmarks. This blurs the boundary between spatial cues and weakens the model's ability to distinguish localized facial features. In contrast, a smaller sigma segments sharper and more localized heatmaps, which better preserve the spatial precision of each landmark. However, this also makes the segmentation more vulnerable to interference from non-target regions such as eyeglasses or background textures. These factors can cause unstable heatmap and degrade the detection results. As shown in Table 2, the detection results consistently peaks when $\sigma = 1$ across all three datasets. This indicates that a moderately balanced spread offers the best trade-off between preserving localized structure and maintaining robustness against noise.

In addition to exploring the impact of pre-training strategies on model performance, we also analyzed how fine-tuning strategies for the heatmap segmentation model affect classification results.

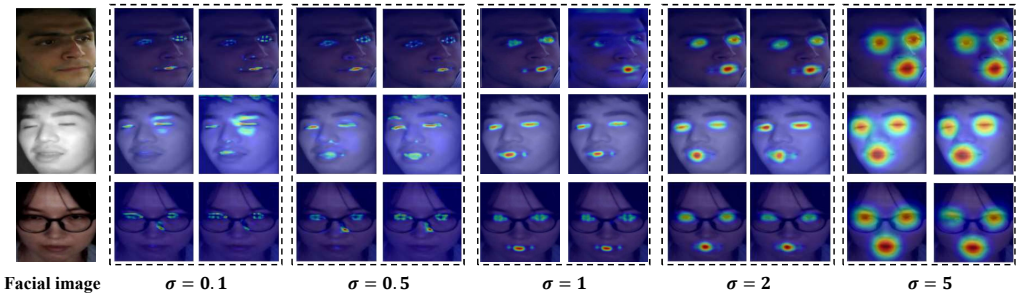


Figure 6: Facial heatmaps before and after training under different Gaussian σ values. Visualization style and annotation conventions are consistent with Fig. 5.

Table 3: Classification performance under different fine-tuning strategies of segmentation model. Ptd. represent pre-trained and Ltd. represent limited learning rate to the segmentation model. The best results for each dataset are marked in bold.

Dataset	Ptd.	Ltd.	Prec.	Rec.	F1	Acc
YawDD[38]		✓	0.7273	0.7031	0.7023	0.7292
			0.9514	0.9456	0.9369	0.9505
	✓		0.9772	0.9739	0.9755	0.9766
	✓	✓	**0.9904**	**0.9935**	**0.9919**	**0.9922**
NTHU-DDD [31]		✓	0.6398	0.6210	0.6011	0.6790
			0.8348	0.8611	0.8405	0.8473
	✓		**0.9524**	**0.9714**	**0.9603**	**0.9630**
	✓	✓	0.9444	0.9331	0.9383	0.9444
VBDDD[34]		✓	0.6168	0.5764	0.5605	0.6250
			0.8114	0.7975	0.7900	0.7976
	✓		0.9068	0.9177	0.9092	0.9108
	✓	✓	**0.9423**	**0.9545**	**0.9456**	**0.9464**

As shown in the table 3, applying limited learning rate to heatmap segmentation model during drowsiness detection training can improve classification result in YawDD[38] and VBDDD[34]. As shown in Fig. 7, the unlimited learning rate of the segmentation model may waken its ability to extract key facial features, affecting training stability. However, since our model is pre-trained on RGB image dataset, its learned representations may not effectively extract facial features from the NTHU-DDD[31] dataset, which

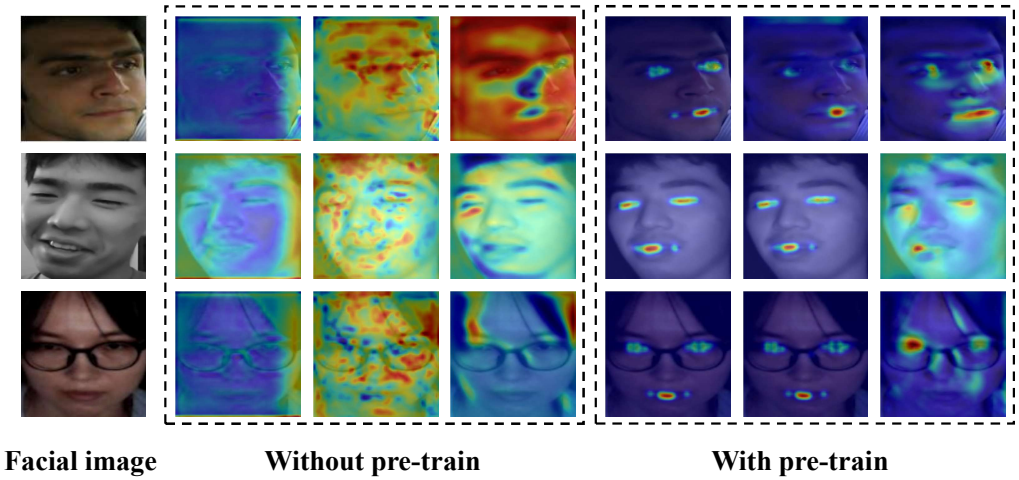


Figure 7: Impact of different training strategies on facial feature extraction. For each column in the figure, the left shows results with pretraining, the middle shows results with a limited learning rate applied to the segmentation model during drowsiness detection training, and the right shows results with an unlimited learning rate applied during training.

consists of grayscale images captured by infrared cameras. Therefore, in experiments on NTHU-DDD[31], a larger learning rate of segmentation model helps compensate for this inadequacies and achieved better classification result.

According to Table 3, results without pre-training are much lower than those with pre-trained models, which highlights the importance of pre-training the segmentation model. As shown in Fig. 7, a pre-trained segmentation model can provide more stable and structural feature representations, thereby improving the downstream classification performance across different datasets.

4.3. Ablation study

To evaluate the effectiveness of our proposed driver drowsiness detection framework, we conducted ablation studies on the above drowsiness-related datasets, comparing our framework with structurally similar ones. The experimental setups are described as follows:

Experiment without pre-stacked heatmaps,to demonstrate the importance of a trainable heatmap segmentation model, we replaced our segmentation module with the FARL[29], a face alignment model,to generate driver facial heatmaps in this experiment, as shown in Fig. 8(a).

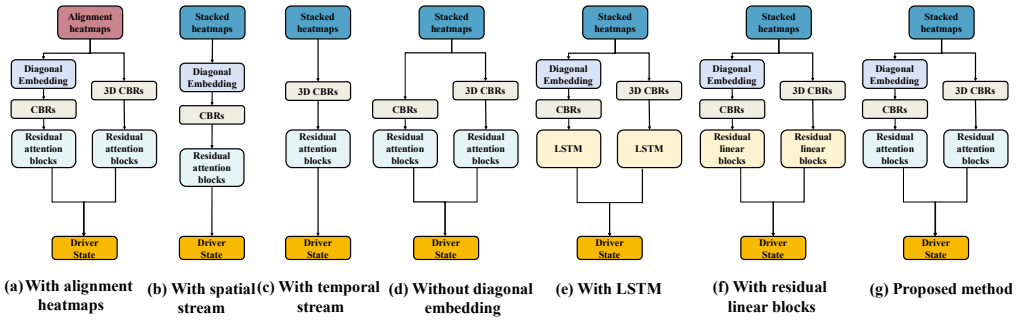


Figure 8: Different model structures of ablation study of this article. Alignment heatmap refers to stacking the outputs of face alignment model before performing regression, in the stacking strategy described in Section 3.

Experiment with single stream,to evaluate the effectiveness of the spatial-temporal two-stream analysis structure, we conducted experiments by separately removing the temporal stream and the spatial stream, as shown in Fig. 8(b) and Fig. 8(c).

Experiment without diagonal embedding,to evaluate the effectiveness of heatmap diagonal embedding, we removed diagonal embedding from the spatial stream in our method to verify whether enhance spatial feature of the heatmaps affects detection result, as shown in Fig. 8(d).

Experiment without residual attention block ,to compare which network structure is more suitable for our analysis module, we replaced the residual attention modules with either residual linear modules or LSTM, in order to explore which performs better within the proposed driver drowsiness detection framework, as shown in Fig. 8(e) and Fig. 8(f).

Experiment with our proposed method, we conduct an experiment with our method as the control group, as shown in Fig. 8(g).

The results of these experiments are shown in table 4. The experimental results on the three driver drowsiness detection datasets validated the effectiveness of our proposed framework. Compared to the heatmap based on face alignment, our trainable segmentation model demonstrates superior result. This may be attributed to the fact that FARL does not stack landmarks features during training. Instead, it segment individual heatmaps for each landmark, and then applies Soft-argmax[28] to obtain the coordinates of the landmarks. As a result, the heat intensity of each heatmap is generated independently and may vary significantly, which can cause some landmarks

Table 4: Performance under different ablation settings across three datasets. The best results for each dataset are marked in bold.

Metric	**Ablation Setting**						
	Alignment	Spatial Stream	Temporal Stream	-Diagonal embedding	LSTM	Residual Linear	Proposed
YawDD[38]							
Prec.	0.8470	0.7933	0.9059	0.9266	0.8234	0.9328	**0.9904**
Rec.	0.7976	0.7716	0.8978	0.9135	0.8097	0.9162	**0.9935**
F1	0.8095	0.7781	0.9013	0.9170	0.8148	0.9162	**0.9919**
Acc	0.8281	0.7943	0.9062	0.9167	0.8255	0.9219	**0.9922**
NTHU-DDD[31]							
Prec.	0.7475	0.8740	0.9081	0.9202	0.5569	0.8952	**0.9444**
Rec.	0.7566	0.8859	0.9097	0.9376	0.6049	0.9085	**0.9331**
F1	0.7410	0.8772	0.9086	0.9272	0.5688	0.9003	**0.9383**
Acc	0.7686	0.8859	0.9167	0.9321	0.6728	0.9105	**0.9444**
VBDDD[34]							
Prec.	0.7506	0.9077	0.8707	0.8705	0.8776	0.9157	**0.9423**
Rec.	0.7474	0.9089	0.8666	0.8808	0.8632	0.9161	**0.9545**
F1	0.7343	0.9099	0.8622	0.8730	0.8686	0.9144	**0.9456**
Acc	0.7500	0.9107	0.8661	0.8750	0.8690	0.9167	**0.9464**

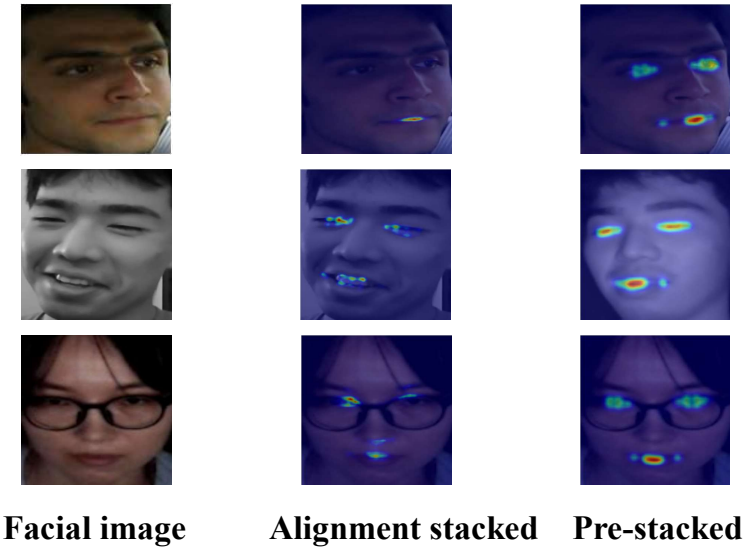


Figure 9: Comparison between pre-stacked segmentation heatmaps and stacked alignment output heatmaps.

to dominate while suppressing the influence of others when the heatmaps are stacked,as shown in Fig. 9.

Removing either the spatial stream or the temporal stream from the

two-stream structure causes a clear drop in results. It can be concluded that both streams contribute essential information to analyze the static and dynamic feature of video-based drowsiness detection. The removal of diagonal embedding also results in lower metrics, indicating that enhance horizontal and vertical features of heatmaps contribute to improve detection results.

Finally, comparisons among different network structure show that the residual attention blocks achieve better results than both the residual linear blocks and LSTM. The LSTM performs worse, possibly due to the limited sequence length after frame sampling, which reduces its capacity to model long-term temporal patterns. Overall, our proposed framework consistently achieves the best results across all datasets, demonstrating its effectiveness in analyzing drowsiness features.

4.4. Comparison

To evaluate the advantages of our proposed framework, we conducted experiments on the above three driver drowsiness detection dataset and compared the results with current state-of-the-art methods.

The experimental results is shown in table 5, which demonstrate that our method achieves the best overall results across all three datasets. Compared with methods such as 2S-STGCN[23] and VBFLLA[34] that use landmarks as facial features, ours avoids the need for external face alignment model to extract landmarks, which requires additional computation time and compromise real-time performance. Similarly, methods like Inception-v3+LSTM[40] and IsoSSL-MoCo[26] rely on LSTM to analyze video sequences, which requires frame-by-frame processing before input to the model, increasing the detection delay.

In contrast, ours uses sampled heatmaps as facial features, enabling fast and accurate drowsiness detection. This makes our framework an efficient and effective solution for real-time driver drowsiness detection. Although our F1 score on the NTHU-DDD[31] dataset is slightly lower than IsoSSL-MoCo[26], previous experiments have shown that ours can outperform it when not trained with limited learning rate. Therefore, we conclude that ours performs excellent in video-based driver drowsiness detection.

4.5. Latency analysis

To evaluate the real-time performance and computational efficiency of the proposed method, we conducted tests on three representative hardware platforms: the NVIDIA GeForce RTX 4090 and RTX 3090 desktop GPUs,

Table 5: Comparison with some existing state-of-the-art methods on three datasets. The best results for each dataset are marked in bold.

Dataset	Method	Facial Features	Feature Analysis Method	Acc	F1
YawDD[38]	CNN-FaceNet[25]	Facial gradient and orientation	Linear neural network	0.9200	–
	3D-LTS[32]	–	3DCNN and BiLSTM	0.9210	–
	2S-STGCN[23]	Landmarks	Spatial and temporal GCN	0.9340	0.895
	HTDBN-HMM[31]	Landmarks	DBN and HMM	0.9400	–
	Inception-v3 +LSTM[40]	–	CNN and LSTM	0.9840	–
	IsoSSL-MoCo[26]	ROI and optical flow	LSTM and Attention	0.9865	0.9837
	VBFLLA[34]	Landmarks and ROI	Spatial and temporal Attention	0.9893	–
	Proposed method(Ours)	Heatmaps	Spatial and temporal Attention	**0.9922**	**0.9919**
NTHU-DDD[31]	HTDBN-HMM[31]	Landmarks	DBN and HMM	0.8482	0.8539
	CNN+TSC-LSTM -Refinement[22]	ROI	3DCNN and LSTM	0.8485	–
	FCFS[14]	–	Temporal Aggregation	0.8763	0.8641
	ResNeXt3D-101 +LSTM[41]	Landmarks	3DCNN and LSTM	0.8860	–
	2S-STGCN[23]	Landmarks	Spatial and temporal GCN	0.9270	0.9260
	IsoSSL-MoCo[26]	ROI and optical flow	LSTM and Attention	0.9371	**0.9464**
	VBFLLA[34]	Landmarks and ROI	Spatial and temporal Attention	0.9310	–
	Proposed method(Ours)	Heatmaps	Spatial and temporal Attention	**0.9444**	0.9383
VBDDD[34]	VBFLLA[34]	Landmarks and ROI	Spatial and temporal Attention	0.9168	0.9093
	Proposed method(Ours)	Heatmaps	Spatial and temporal Attention	**0.9464**	**0.9456**

the NVIDIA GeForce GTX 1650 Laptop GPU, and the NVIDIA Jetson AGX Orin as a representative in-vehicle embedded platform, as it is widely adopted in current intelligent driving systems and considered a mainstream automotive-grade AI SoC[42]. We compared three key metrics during inference: memory cost peak, inference delay, and total delay per video. The experimental results shown in table 6 confirm that our method demonstrates excellent real-time performance across all computing platforms.

Notably, the total delay is consistently higher than the inference time, primarily due to the time-consuming face detection step across T frames, which is the main contributor to increased latency in the driver drowsiness detection pipeline.

Our method demonstrates consistent efficiency across various platforms, including desktop GPUs, laptop GPU and in-vehicle embedded platform.

Table 6: The memory cost and time delay during the proposed pipeline per video on different hardware platforms. Inference refers to the process performed by the facial feature extraction module and analysis module.

Device	Inference Cost (MB)	Inference Delay (ms)	Total Delay (ms)
RTX 4090	2480	146.97	393.38
RTX 3090	2294	344.61	572.14
GTX 1650 Laptop	3689	788.54	1740.18
AGX Orin	6775	311.80	596.92

Table 7: Comparison of total processing delay with some existing video-based drowsiness detection methods.

Method	Device	Total Delay (ms)
FCFS[14]	2 × RTX 2080	849
ResNeXt3D-101 +LSTM[41]	2 × Tesla K80	–
2S-STGCN[23]	GTX 1080	–
VBFLLA[34]	2 × RTX 3090	790
Ours	RTX 3090	572.14
Ours	AGX Orin	596.92

Despite the AGX Orin exhibiting the highest memory cost peak due to its unified memory architecture, it achieves inference and total delays comparable to those of the RTX 3090. This highlights the method's strong potential for deployment in real-world automotive embedded systems. Furthermore, although latency increases on the lower-end GTX 1650 Laptop GPU, the method remains functional, demonstrating its scalability across devices with varying computational capacities.

To further assess the real-time capability of our approach, we compared its total delay against several existing video-based driver drowsiness detection methods under their reported hardware settings. As shown in Table 7, although a direct comparison of inference delays across different hardware platforms may not be fully rigorous, the results still provide valuable insights into the relative computational efficiency.

While cross-platform comparisons are not strictly equivalent, the results in Table 7 indicate that our method achieves lower processing latency with

significantly lower hardware requirements. This highlights its superior computational efficiency and potential for real-world deployment on embedded platforms.

5. Conclusion

In this article,we proposed a video-based driver drowsiness detection framework using stacked landmark heatmaps and spatial-temporal attention for efficient and accurate drowsiness detection. By replacing face alignment model with a trainable heatmap segmentation model and then applied limited fine-tuning specifically for drowsiness detection, our method reduces cumulative errors while enabling task-specific feature learning. The diagonal embedding encodes horizontal and vertical geometric relationships between landmarks, enhancing spatial representation in low-resolution heatmaps. Through a two-stream attention structure combining 2D and 3D convolutions, the framework achieves spatial-temporal analysis with real-time inference. Extensive experimental validations demonstrate the framework's superior performance across public benchmarks, achieving leading accuracy in drowsiness detection tasks. Latency analysis on hardware platforms further confirm the method's computational efficiency, and the potential of real-world deployment.

However, since our method is a vision-based detection method, it remains vulnerable to failure under conditions where facial features are partially lost or distorted. In particular, occlusions caused by sunglasses or masks, as well as motion-induced blur from sudden head movements, can negatively impact the quality of heatmap segmentation, thereby affecting the accuracy of drowsiness detection. While the method demonstrates reasonable robustness to transparent eyeglasses, these more severe occlusion or distortion scenarios still present challenges. Future work may consider incorporating physiological signals to enable more reliable drowsiness detection under such visually compromised conditions.

Moreover, our heatmap segmentation remains dependent on face detection, which introduces latency bottlenecks during drowsiness detection. This dependency limits further improvements in end-to-end real-time performance. Future work will explore unified architectures that integrate data preprocessing and facial feature extraction to achieve truly end-to-end video-based driver drowsiness detection.

Conflicts of Interest

The authors declared that they have no conflicts of interest to this work.

Data availability Statement

All data included in this study are available upon request by contact with the corresponding author.

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